# Overview of Acyclovir Pharmacokinetic Disposition in Adults and Children

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**The metabolic disposition and pharmacokinetics of acyclovir have been studied as part of the clinical evaluation of the drug in humans. Data from 10 studies have been summarized and, when appropriate, pooled across studies for further analysis. The principal findings are as follows: Renal excretion is the major route of elimination of acyclovir and is dependent, in part, on active tubular secretion. Total body clearance ($Cl_{tot}$) and half-life are dependent on renal function as evaluated by estimated creatinine clearance ($Cl_{cr}$). $Cl_{tot}$ is markedly reduced in the anuric patient. Plasma protein binding is low and drug interactions involving binding displacement are not anticipated. Acyclovir levels in cerebrospinal fluid are approximately 50 percent of corresponding plasma levels. Dose-independent pharmacokinetics is observed in the range of 0.5 to 15 mg/kg. Proportionality between dose and plasma levels is seen after single doses or at steady state after multiple dosing. Similar plasma levels are achieved in adults and pediatric patients (>1 year) when equivalent doses are given based on body surface area. Intrasubject variability of acyclovir disposition is low. Much but not all intersubject variability in $Cl_{tot}$ can be explained by differences in renal function. Dosage adjustments for various stages of renal impairment are proposed based on the observed relationship between $Cl_{tot}$ and $Cl_{cr}$.**

Clinical evaluation of a new therapeutic agent is greatly aided by the early definition of its pharmacokinetic behavior. The pharmacokinetic parameters obtained from initial single-dose studies provide the information needed to optimize multiple dosing regimens in efficacy trials. Pharmacokinetic data obtained during efficacy evaluations can also aid in the establishment of rational dosing guidelines for the drug's general clinical use. Finally, continued pharmacokinetic studies during clinical trials can lead to recognition and evaluation of drug interaction and the influence of disease states on drug absorption and disposition.

The metabolic disposition and pharmacokinetics of acyclovir in humans have been extensively investigated in several Phase I and II safety and efficacy trials. The principal studies are summarized in **Table I**. The pharmacokinetics of acyclovir in adults has been evaluated after single doses ranging from 0.5 to 15 mg/kg (Study 2, 3, and 4) and after multiple doses ranging from 2.5 to 15 mg/kg three times daily (Study 5 and 16). In all studies, acyclovir was administered as a one hour infusion except in Study 3 where the pharmacokinetics was compared after one hour and six hour infusions. The metabolic fate of acyclovir was investigated after the administration of $^{14}C$-labeled

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**TABLE I Pharmacokinetic Studies of Acyclovir (One-Hour Intravenous Infusions)**

| Study No. | Principal Investigator | Condition of Patients | Dosage (mg/kg) | Type of Study | No. Doses | No. Patients (male/female) | Age (yrs) (mean ± SD) |
|---|---|---|---|---|---|---|---|
| 2 | a | Non-herpes-infected advanced cancer | 0.5, 1.0, 2.5, 5.0 | Single-dose pharmacokinetics | 1 | 10/4 | 61 ± 10 |
| 3* | b | Non-herpes-infected advanced cancer | 0.5, 1.0 | Single-dose pharmacokinetics | 1 | 7/3 | 60 ± 14 |
| 4 | c | Non-herpes-infected advanced cancer | 2.5, 5.0, 10, 15 | Single-dose pharmacokinetics | 1 | 6/8 | 58 ± 14 |
| 5 | a | Non-herpes-infected advanced cancer | 2.5, 5.0, 10 every 8 hours | Multiple-dose pharmacokinetics | 3 | 9/0 | 61 ± 9 |
| 16 | a | Immunocompromised herpes-infected | 2.5, 5.0, 10, 15 every 8 hours | Multiple-dose pharmacokinetics | 16 | 6/14 | 45 ± 17 |
| 22 | d | Pediatric herpes-infected | 250,† 500† every 8 hours | Multiple-dose pharmacokinetics | 15 | 15/5 | ‡ |
| 23 | d | Neonatal herpes-infected | 5.0, 10, 15 every 8 hours | Multiple-dose pharmacokinetics | 16 | 5/6 | 10 ± 7§ |
| 4A | c | Non-herpes-infected advanced cancer | 5.0 | Probenecid interaction | 1 on two occasions | 1/2 | 47 ± 22 |
| 3A/4B | b, c | Non-herpes-infected advanced cancer | 0.5, 2.5 | 14-C metabolism | 1 | 1/4 | 53 ± 18 |
| 4C | c | Chronic renal failure non-herpes-infected | 2.5 | Chronic renal failure hemodialysis | 1 | 4/2 | 46 ± 18 |

* Drug given as one- and six-hour infusions.
† Dose/$m^2$ (roughly equivalent to 6 and 12 mg/kg).
‡ See Table IV for age distribution.
§ Age in days at start of therapy.
a = Dr. R. Whitley, University of Alabama, Birmingham, Alabama.
b = Dr. J. Connor, University of California, San Diego, California.
c = Dr. P. Lietman, Johns Hopkins University, Baltimore, Maryland.
d = Multicenter study.

drug (Study 3A/4B). The pharmacokinetics in neonates (Study 23) and in a pediatric population aged 3 months to 17 years (Study 22) has also been evaluated as has been the kinetics in patients with chronic renal failure and the effects of hemodialysis on drug removal (Study 4C). The influence of probenecid on acyclovir elimination has also been investigated (Study 4A).

## PATIENTS AND METHODS

**Patient Population. Adults:** For the initial Phase I investigations, advanced cancer patients who were at high risk but not infected with herpes virus were selected. Patients with renal or hepatic impairment were to be excluded. The mean age for the non-herpes-infected patients was approximately 60 years. For Phase II studies herpes-infected patients were selected and the mean age for these immunocompromised patients was 45 years (Study 16). All patients gave informed consent.

In these pharmacokinetic studies all patients received some concurrent medications. Although patients with normal renal function were to be selected, individual estimated creatinine clearance ($Cl_{cr}$) values varied considerably within studies and mean $Cl_{cr}$ values also varied between studies (**Table II**).

**Children:** All patients were being treated for herpes infections and, for pharmacokinetic analysis, were divided into age ranges of birth to three months (neonates), three months to one year, one to two years, two to seven years, seven to 12 years, and 12 to 17 years.

**Drug Administration.** For all intravenous studies the lyophilized sodium salt of acyclovir was reconstituted to a 20 mg/ml solution with sterile water for injection. Calculated unit doses were added to lactated Ringer's solution to bring the total volume to 60 ml. Doses at the level of 15 mg/kg were diluted to a total volume of 120 ml. Drug solution was infused over one hour at a constant rate into a peripheral vein.

**Assay Methods.** Levels of acyclovir in plasma, urine, and cerebrospinal fluid were measured by a sensitive and specific radioimmunoassay previously described [1]. The assay requires 100 $\mu$l or less of biologic fluid and has a lower limit of sensitivity from 1.1 to 11.3 ng/ml (conversion factor: 1 $\mu M$ = 0.225 $\mu$g/ml) depending on the antiserum batch used. Samples were run in triplicate. All analyses for studies listed in Table I were performed by the Experimental Therapy Department, Wellcome Research Laboratories, with the exception of Study 22. The assays for some patients in this study were run in the Acyclovir Radioimmunoassay Laboratory, University of California, San Diego. Quality control was maintained between the two laboratories.

**Pharmacokinetic Analysis Methods.** Plasma concentration–time curves exhibited a biexponential decay after the end of infusion. Therefore, individual patient data were analyzed through the use of the nonlinear least-squares regression program NONLIN [2] in conjunction with a specific subroutine for the two-compartment open pharmacokinetic

**TABLE II Principal Pharmacokinetic Parameters From Adult Studies***

| Study No. | $\alpha$ | $\beta$ | $t_{1/2\beta}$ (hr) | $V_1$ | $Vd_\beta$ | $Vd_{ss}$ | $Cl_{tot}$ | $Cl_r$ | $Cl_{cr}$ | Recovery† % |
|---|---|---|---|---|---|---|---|---|---|---|
| | $hr^{-1}$ | | | liters/1.73 $m^2$ | | | | ml/min/1.73 $m^2$ | | |
| 02 | 2.61 ± 1.50 | 0.231 ± 0.054 | 3.16 ± 0.76 | 22.2 ± 15.8 | 55.9 ± 26.4 | 45.0 ± 20.1 | 201 ± 63 | 125 ± 54 | 57 ± 13 | 56 ± 12 |
| 03 | 2.21 ± 1.03 | 0.256 ± 0.036 | 2.76 ± 0.47 | 25.9 ± 10.3 | 74.1 ± 18.6 | 55.0 ± 13.2 | 303 ± 52 | 179 ± 52 | 83 ± 35 | 49 ± 9 |
| 04 | 2.53 ± 1.00 | 0.273 ± 0.054 | 2.63 ± 0.52 | 18.3 ± 3.5 | 66.1 ± 11.6 | 46.6 ± 8.5 | 292 ± 82 | 234 ± 100 | 71 ± 31 | 70 ± 15 |
| 05 | 2.41 ± 0.90 | 0.266 ± 0.071 | 2.78 ± 0.73 | 17.9 ± 6.5 | 55.1 ± 12.9 | 42.5 ± 5.9 | 249 ± 101 | 177 ± 87 | 67 ± 25 | 60 ± 12 |
| 16 | 2.36 ± 2.01 | 0.222 ± 0.069 | 3.43 ± 1.13 | 21.2 ± 8.4 | 72.2 ± 19.4 | 50.2 ± 12.7 | 263 ± 90 | 184 ± 79 | 70 ± 41 | 68 ± 14 |
| 3A/4B | n.d. | 0.355 ± 0.080 | 2.05 ± 0.52 | n.d. | n.d. | n.d. | 297 ± 53 | 240 ± 77 | 80 ± 28 | 75 ± 17 |
| 4A‡ | 2.70 ± 0.53 | 0.304 ± 0.034 | 2.30 ± 0.26 | 17.1 ± 1.8 | 58.9 ± 9.5 | 41.4 ± 1.7 | 300 ± 69 | 248 ± 80 | 90 ± 48 | 79 ± 9 |
| Overall mean | 2.45 ± 1.41 | 0.256 ± 0.068 | 2.91 ± 0.85 | 21.1 ± 10.0 | 65.0 ± 19.8 | 48.0 ± 13.6 | 260 ± 81 | 186 ± 82 | 71 ± 32 | 63 ± 15 |
| (Range) | (0.68 −9.01) | (0.110 − 0.456) | (1.52 − 6.29) | (6.3 − 55.7) | (28.7 − 131) | (22.5 − 101) | (91 − 504) | (47 − 445) | (19 − 220) | (30 − 96) |

* Mean ± SD.
† 72 hour urinary excretion of unchanged acyclovir.
‡ Data from control portion of study (no probenecid).
n.d. = Not determined.

model with zero-order input [3]. When appropriate, pharmacokinetic parameters were normalized according to body surface area. Acyclovir renal clearance ($Cl_r$) was determined from plasma and urine data [3] and $Cl_{cr}$ were estimated according to the method of Mawer et al. [4].

## RESULTS AND COMMENTS

**Elimination Processes. Renal excretion:** The major route of acyclovir elimination is by excretion into the urine. In the various studies, fractional urine collections were made for 72 hours and analyzed for unchanged drug. Generally 90 to 95 percent of the ultimate urinary recovery of acyclovir is excreted during the first 12 hours after the start of infusion. The mean percent acyclovir urinary recovery for the various studies is presented in Table II. The average recovery across all studies was 63 ± 15 percent. The highest mean recovery of 79 percent was in Study 4A (n = 3). The volunteers in this study also had the best renal function as indicated by highest mean $Cl_{Cr}$ (90 ml/min per 1.73 $m^2$). An apparent correlation between mean $Cl_{cr}$ and acyclovir urinary recovery (with the exception of Study 3) can be seen from the data in Table II. This relationship is expected since, as renal function is decreased, a greater percent of the drug will be eliminated by nonrenal pathways. In all studies renal clearance of acyclovir was substantially higher than estimated creatinine clearance, indicating that not only glomerular filtration but also tubular secretion contributes to the renal elimination of the drug.

Because the excretion of acyclovir was found to be dependent on active tubular secretion, a study (4A) was carried out to evaluate the influence of probenecid on the pharmacokinetics and renal clearance of acyclovir. In a two-way crossover study in three volunteers with advanced malignancies (normal renal function), acyclovir 5 mg/kg was administered by one-hour intravenous infusion. In one leg of the crossover the patients received only the acyclovir and in the other acyclovir was given one hour after 1 g oral probenecid. After the probenecid treatment, the acyclovir half-life ($t_{1/2\beta}$) and area under the plasma concentration–time curve increased by 18 and 40 percent, respectively. The mean urinary recovery of acyclovir declined from 79 to 69 percent of dose and renal clearance ($Cl_r$) decreased from 248 to 168 ml/min/1.73 $m^2$. Although renal clearance dropped with concomitant probenecid administration, it was still approximately twice the value of estimated creatinine clearance. In patients on chronic probenecid therapy, the influence of this drug on acyclovir elimination might be more significant.

**Metabolic elimination:** The metabolic disposition of acyclovir was studied in five volunteers receiving a single dose of 0.5 or 2.5 mg/kg $^{14}$C-acyclovir (labeled in the 8 position of the purine ring) by one-hour infusion

[5]. Each subject received a tracer dose of 100 μCi of drug. Blood, plasma, urine, feces, and expired $CO_2$ were collected. Most of the radioactive dose was recovered in the urine (71 to 99 percent) with less than 2 percent excretion in the feces and only trace amounts in expired $CO_2$. Analyses of the various urine samples by reverse-phase high-performance liquid chromatography (HPLC) and further characterization by thin-layer chromatography indicated that 9-carboxymethoxymethylguanine (CMMG) was the only significant metabolite of acyclovir, accounting for 8.5 to 14.1 percent of the dose. A minor metabolite (<0.15 percent of dose) had the retention time of 8-hydroxy-9-(2-hydroxyethoxymethyl)guanine. There was no indication of acyclovir cleavage to guanine.

High-performance liquid chromatographic analysis was used to quantitate CMMG urinary excretion from two patients in Study 16 who had moderately impaired renal function and low unchanged acyclovir urinary recovery. Twenty-one and 44 percent of the dose was excreted as CMMG in these patients who had $Cl_{cr}$ of 53 and 18 ml/min/1.73 $m^2$, respectively, indicating that more drug is eliminated by metabolism as renal function decreases. CMMG levels have not been determined in plasma but the disposition kinetics of total radioactivity is nearly identical to acyclovir measured by radioimmunoassay, indicating no metabolite accumulation. Also greater than 99 percent of the radioactivity excreted over 72 hours is excreted during the first 24 hours, again indicating no drug or metabolite retention [5]. CMMG has little antiherpetic activity.

**Plasma Protein Binding.** Measurements of acyclovir binding to plasma proteins were carried out by ultrafiltration of 12 plasma samples of four patients given $^{14}C$-acyclovir (Study 03A/04B). Plasma binding for the various samples (acyclovir concentrations of 0.4 to 5.0 μg/ml) ranged from 9 to 22 percent with a mean value of 15.4 ± 4.4 percent.

In another study [3] in which $^{14}C$-acyclovir was incubated with normal human plasma at 37°C, binding of 22 and 33 percent was observed at acyclovir concentrations of 4.0 and 0.4 μg/ml, respectively. Because of relatively low plasma protein binding, acyclovir is not expected to cause or be subject to drug interactions mediated by protein binding displacement.

**Drug Plasma Concentrations. Single dose in adults:** Proportionality between dose and plasma concentrations in adult patients was observed in the various single-dose studies. Mean peak (end of infusion) acyclovir concentrations pooled across single-dose one-hour administrations, regardless of renal function, were 1.0 ± 0.4, 2.1 ± 1.0, 4.2 ± 0.8, 8.8 ± 3.8, 14.6 ± 2.0, and 22.7 ± 10.4 μg/ml for doses of 0.5, 1.0, 2.5, 5.0, 10.0, and 15.0 mg/kg, respectively. The high intersubject variability for some dosage levels, indicated by large standard deviations about the means, generally reflects a high variability in renal function among the patients at that dose level.

**TABLE III Steady State Acyclovir Plasma Levels After One-Hour Infusions Every Eight Hours in Adults**

| Dose (mg/kg) every 8 hr | Number of Patients | Plasma Concentration (μg/ml)* $C^{ss}_{max}$ | $C^{ss}_{min}$ |
|---|---|---|---|
| 2.5† | 5 | 5.1 ± 1.0 | 0.5 ± 0.5 |
| 5.0 | 8 | 9.8 ± 2.6 | 0.7 ± 0.3 |
| 10.0 | 9 | 20.7 ± 10.2 | 2.3 ± 1.4 |
| 15.0‡ | 2 | 23.6 ± 0.2 | 2.0 ± 0.1 |

* Mean ± SD.
† Data from patient with creatinine clearance of 18 ml/min/1.73 $m^2$ excluded from mean.
‡ Data from patient with creatinine clearance of 220 ml/min/1.73 $m^2$ excluded from mean.

**Multiple dose in adults:** In the adult multiple dosing studies proportionality between steady state plasma concentrations and dosage was also observed. Mean steady state peak concentrations ($C^{ss}_{max}$) for the combined data were 5.1 ± 1.0 (n = 5), 9.8 ± 2.6 (n = 8), 20.7 ± 10.2 (n = 9), and 23.6 ± 0.2 μg/ml (n = 2), respectively, for doses of 2.5, 5.0, 10, and 15 mg/kg every eight hours. In determining mean values, two outlying observations were omitted that were associated with the very high and low creatinine clearance values of 220 and 18.9 ml/min/1.73 $m^2$, respectively. A linear correlation of $C^{ss}_{max}$ and dose is apparent (r = 0.93) from the linear regression analysis forced through the origin. Peak levels after multiple dosing are similar to peak levels after single equivalent doses, indicating little drug accumulation when patients are dosed every eight hours.

The mean steady state minimum concentrations ($C^{ss}_{min}$) for the combined multiple dose studies, minus the two outlyers noted above, were 0.5 ± 0.5, 0.7 ± 0.3, 2.3 ± 1.4, and 2.0 ± 0.1 μg/ml, respectively, for the 2.5, 5.0, 10.0, and 15 mg/kg given every eight hours. The steady state maximum and minimum acyclovir levels are summarized in **Table III**.

**Multiple dose in pediatric patients:** Although there is considerable intersubject variability in some age groups, all patients given 250 mg/$m^2$ and 500 mg/$m^2$ achieved peak steady state concentrations between 4 and 18 μg/ml and between 14 and 30 μg/ml, respectively. The overall mean $C^{ss}_{max}$ of 10.3 ± 4.0 and 20.7 ± 5.0 μg/ml for the 250 and 500 mg/$m^2$ regimens were nearly identical to the mean $C^{ss}_{max}$ in adults receiving nearly equivalent doses of 5 and 10 mg/kg every eight hours (Table III). Good correspondence in $C^{ss}_{min}$ values were also seen between equivalent doses in children and adults.

**TABLE IV Principal Pharmacokinetic Parameters in Children***

| Age Group | Number of Patients | $Cl_{tot}$ (ml/min/1.73 $m^2$) | $t_{1/2\beta}$ (hr) | $Vd_{ss}$ (liters/1.73 $m^2$) |
|---|---|---|---|---|
| 0–3 months† | 11 | 105 ± 42 | 4.05 ± 1.22 | 28.8 ± 9.3 |
| 3 mo–1 yr† | 0 | — | — | — |
| 1–2 yr | 4 | 325 ± 76 | 1.86 ± 0.42 | 31.6 ± 4.2 |
| 2–7 yr | 5 | 366 ± 101 | 2.16 ± 1.08 | 42.0 ± 13.0 |
| 7–12 yr | 6 | 353 ± 142 | 2.81 ± 1.10 | 51.2 ± 18.3 |
| 12–17 yr | 3 | 263 ± 95 | 3.58 ± 0.59 | 53.6 ± 14.6 |
| Overall mean | 18 | 335 ± 109 | 2.52 ± 1.04 | 44.7 ± 15.4 |

* Mean ± SD.
† Excluded from overall mean.

**Distribution Into Cerebrospinal Fluid (CSF).** In Study 16, samples of CSF were obtained from two patients receiving 10 mg/kg three times daily and analyzed for acyclovir by radioimmunoassay. Both samples were collected two hours after the end of the infusion. Although corresponding plasma samples were not obtained, levels of acyclovir in CSF of 3.7 and 4.0 μg/ml can be compared with model-predicted plasma levels of 7.9 and 8.1 μg/ml, respectively, at the times of CSF sampling. In the pediatric study a CSF level of 4.0 μg/ml with a corresponding predicted plasma level of 9.7 μg/ml was determined in one patient. These data indicate that acyclovir penetrates the blood–cerebrospinal fluid barrier and CSF levels are approximately 50 percent of corresponding plasma levels.

**Pharmacokinetic Analysis. Pharmacokinetic parameters in adults:** Plasma acyclovir concentration data from all single-dose and multiple-dose studies were well fitted by a two-compartment open model with zero-order input [6]. The principal pharmacokinetic parameters in adult patients are summarized according to study in Table II. Since initial evaluation indicated no differences in parameters from single-dose or multiple-dose studies, the data from the various studies were pooled to determine overall means. The overall $t_{1/2\beta}$ is 2.91 ± 0.85 hour (range, 1.5 to 6.3), and $Cl_{tot}$ is 260 ± 81 ml/min/1.73 $m^2$ (range, 91 to 504). The mean volume of distribution of the central compartment, $V_1$ (i.e., blood and highly perfused tissues), across all studies is 21.1 ± 10.0 liters/1.73 $m^2$. The volume of distribution beta, $Vd_\beta$, is 65.0 ± 19.8 liters/1.73 $m^2$ and, as expected, is dependent on the slow disposition rate constant, $\beta$ (compare with Table V). The most meaningful volume term, volume of distribution steady state, $Vd_{ss}$, has an overall mean value of 48.0 ± 13.6 liters/1.73 $m^2$ and is the proportionality constant relating plasma concentration to amount of drug in the body at steady state conditions. The values of $Vd_\beta$ and $Vd_{ss}$ indicate acyclovir is taken up into tissue.

There is no apparent difference in the pharmacokinetic disposition of acyclovir in patients with or without herpes infections or between males and females. As anticipated, the length of infusion (one hour versus six hours) did not influence the pharmacokinetic disposition of acyclovir. In a study carried out in the Wellcome Research Laboratories at Beckenham, England [7], intravenous bolus doses of 50 mg (mean doses of 0.73 mg/kg) were given to six healthy volunteers. The mean $Cl_{tot}$, $Cl_r$, $t_{1/2\beta}$, and $Vd_\beta$ values were 254 ± 50 ml/min, 185 ± 37 ml/min, 2.88 ± 0.50 hour, and 62.6 ± 13.4 liters. These values were nearly identical to the corresponding mean values across all the one-hour infusion studies (Table II).

**Pharmacokinetic parameters in pediatric and neonatal patients:** The principal acyclovir parameters derived from the pediatric study are presented, according to age groups, in **Table IV**. No patients were aged three months to one year. In general the pharmacokinetic parameters in the pediatric age groups are comparable to the adult population. The mean $Cl_{tot}$, which reflects the overall elimination of the drug from the body and determines steady state levels in the body after multiple dosing, is similar to the $Cl_{tot}$ in adults with $Cl_{cr}$ >80 ml/min/1.73 $m^2$ (**Table V**). The mean $Cl_{tot}$ (105 ± 42 ml/min/1.73 $m^2$) for 11 neonatal patients, was approximately one-third that of normal adults. This is consistent with known changes in renal function with age [8]. The mean $t_{1/2\beta}$ for these neonates was 4.1 ± 1.2 hours.

**Intrasubject variability:** Information on intrasubject variability of acyclovir pharmacokinetic disposition was gained from two sources. First, the ability to model-fit multiple dose plasma concentration–time data is evidence of consistent pharmacokinetics over the course of treatment. Second, three patients have been evaluated in two or three separate single dose protocols over about a year. They received a different dose each time. The pharmacokinetic parameters for a patient were consistent over time and at different dosages. These data also support dose-independent kinetics over the anticipated clinical dosage range.

**Influence of Impaired Renal Function on Acyclovir Disposition. Relationship between creatinine clearance and pharmacokinetics:** The data on pharmacokinetic parameters represented as mean values

**TABLE V** **Principal Pharmacokinetic Parameters in Adults According to Creatinine Clearance Range***

| Parameter | Creatinine Clearance Range (ml/min/1.73 $m^2$) 15-50 (n† = 12) | 50-80 (n = 38) | >80 (n = 20) |
|---|---|---|---|
| $Cl_{cr}$ (ml/min/1.73 $m^2$) | 38.2 ± 10.0 | 62.9 ± 8.4 | 109 ± 36 |
| $S_{cr}$ (mg/dl) | 1.54 ± 0.58 | 1.07 ± 0.23 | 0.80 ± 0.17 |
| $Cl_{tot}$ (ml/min/1.73 $m^2$) | 190 ± 62 | 248 ± 62 | 327 ± 80 |
| $Cl_r$ (ml/min/1.73 $m^2$) | 122 ± 52 | 175 ± 64 | 248 ± 90 |
| $\alpha$ ($hr^{-1}$) | 2.19 ± 0.61 | 2.36 ± 1.71 | 2.75 ± 1.01 |
| $\beta$ ($hr^{-1}$) | 0.214 ± 0.056 | 0.250 ± 0.062 | 0.295 ± 0.07 |
| $t_{1/2\beta}$ (hr) | 3.48 ± 1.06 | 2.95 ± 0.78 | 2.48 ± 0.64 |
| $V_1$ (liters/1.73 $m^2$) | 21.0 ± 7.8 | 22.2 ± 11.3 | 19.2 ± 8.4 |
| $Vd_\beta$ (liters/1.73 $m^2$) | 55.0 ± 13.2 | 64.3 ± 20.8 | 72.3 ± 18.9 |
| $Vd_{ss}$ (liters/1.73 $m^2$) | 46.6 ± 10.0 | 47.5 ± 14.2 | 49.7 ± 14.5 |
| Recovery (%)‡ | 54.3 ± 13.6 | 64.5 ± 14.1 | 67.0 ± 15.3 |

* Mean ± SD.
† n is based on number of $Cl_{tot}$ observations, n for other parameters may vary slightly.
‡ 72 hour urinary excretion of unchanged acyclovir.

for each study (Table II) were pooled and further analyzed. A multiple linear regression model [9] was used to evaluate the influence of dose and renal function (as indicated by estimated creatinine clearance) on $Cl_{tot}$. The multiple regression coefficient (r) was 0.664 with both $Cl_{cr}$ and dose as dependent variables and was 0.661 with $Cl_{cr}$ alone. It appears that dose has no significant influence on acyclovir clearance.

**TABLE VI** **Dosage Adjustment Recommended for Patients with Renal Impairment**

| Creatinine Clearance (ml/min/1.73 $m^2$) | Percent Usual Dose | Dosing Interval (hours) |
|---|---|---|
| >50 | 100 | 8 |
| 25–50 | 100 | 12 |
| 10–25 | 100 | 24 |
| 0–10 | 50 | 24 |

Because total body clearance was shown to be dependent on renal function, the principal parameters were stratified according to estimated creatinine clearance similar to the method of Bennett [10]. Data were grouped as $Cl_{cr}$ >80 ml/min/1.73 $m^2$ (normal renal function), 50 to 80 ml/min/1.73 $m^2$ (mild renal impairment), and 15 to 50 ml/min/1.73 $m^2$ (moderate renal impairment). The results are presented in Table V. The three groups had a mean $Cl_{tot}$ of 327 ± 80, 248 ± 62, and 190 ± 62 ml/min/1.73 $m^2$ and a corresponding mean $Cl_{cr}$ of 109 ± 36, 62.9 ± 8.4, and 38.2 ± 10.0 ml/min/1.73 $m^2$, respectively. The $t_{1/2\beta}$ for the groups were 2.48 ± 0.64, 2.95 ± 0.78, and 3.48 ± 1.06 hours. The influence of renal function on acyclovir clearance is obvious from these data. The $Vd_{ss}$ was constant in all three groups and the $Vd_\beta$, as anticipated, was dependent on $\beta$. Less drug is recovered unchanged in the low renal function group than in the intermediate and high renal function groups.

The apparent linear relationship between $Cl_{tot}$ and $Cl_{cr}$ was explored further. **Figure 1** shows a plot of $Cl_{tot}$ versus $Cl_{cr}$ from data pooled across all adult studies.

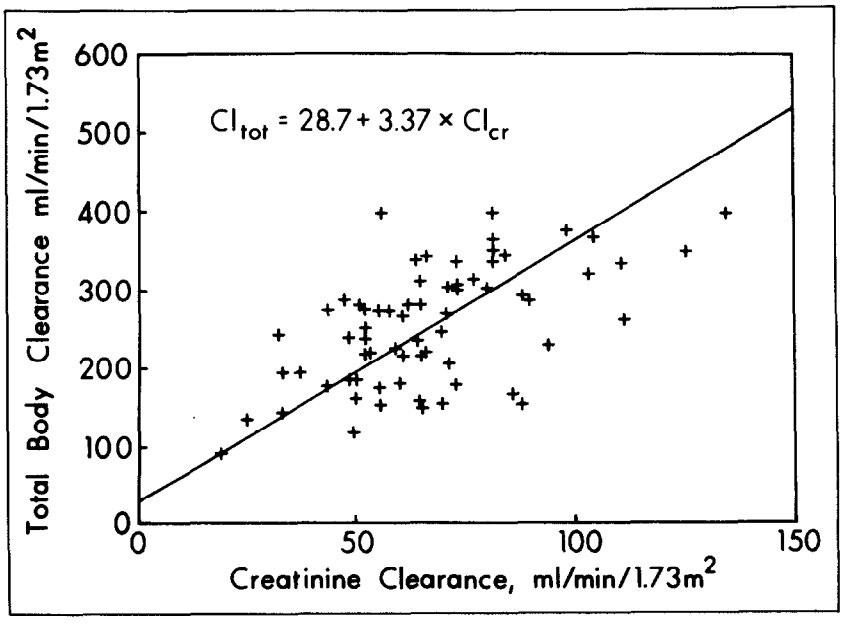



***Figure 1.*** *Total body clearance versus creatinine clearance for acyclovir in adult patients. + = individual observations;* ***solid line*** *= orthogonal regression line described by the equation* $Cl_{tot} = 28.7 + 3.37 \times Cl_{cr}$.

Due to experimental error in determining both parameters, $Cl_{tot}$ and $Cl_{cr}$ were treated as random parameters. The least-squares regression is bivariate and the best functional relationship between the two parameters is described by the orthogonal regression line [11]. When data with $Cl_{cr} > 145$ ml/min/1.73 $m^2$ (three points out of 70) were dropped from analysis, orthogonal regression led to the relationship $Cl_{tot} = 28.7 + 3.37 \times Cl_{cr}$ ($r = 0.58$), which closely predicts the mean $Cl_{tot}$ of the mild and moderate renal impairment groups and also most of the individual values of $Cl_{tot}$ when $Cl_{cr}$ is below 25 ml/min/1.73 $m^2$. The intercept value of this equation, 28.7 ml/min/1.73 $m^2$ is the predicted $Cl_{tot}$ for anuric patients and is nearly identical to the value reported by Laskin et al. [12].

**Pharmacokinetics in chronic renal failure patients and effect of hemodialysis on acyclovir removal:** The pharmacokinetics of acyclovir was studied in six non-herpes-infected volunteers with chronic renal failure (CRF) who were anuric (Study 4C; [12]). Each subject received 2.5 mg/kg acyclovir as a one-hour infusion. After approximately 48 hours the subjects underwent a six-hour hemodialysis treatment using a hollow fiber single-pass system with a blood flow rate of 200 ml/min and a dialysate flow of 800 ml/min. The pharmacokinetic disposition of acyclovir was characterized for the period before and during hemodialysis.

Before the initiation of dialysis the mean acyclovir $t_{1/2\beta}$ and $Cl_{tot}$ in the CRF patients were 19.5 ± 5.9 hours and 28.6 ± 9.5 ml/min/1.73 $m^2$. The mean peak plasma level at the end of infusion was 8.4 ± 5.4 μg/ml and the level dropped to 0.7 ± 0.3 μg/ml at approximately 48 hours. The mean acyclovir $t_{1/2\beta}$ and $Cl_{tot}$ during dialysis were 5.7 ± 0.9 hours and 113 ± 16 ml/min/1.73 $m^2$, respectively. The calculated extraction ratio was 0.44. Plasma levels of acyclovir dropped approximately 60 percent during the course of dialysis. Hemodialysis is therefore effective in removing acyclovir from the body and replacement doses should be considered in patients undergoing dialysis.

**Dosage Recommendations in Renal Impairment.** Guidelines for dosing adjustments have been developed using the derived relationship between $Cl_{tot}$ and $Cl_{cr}$ ($Cl_{tot} = 28.7 + 3.37 \times Cl_{cr}$). For various creatinine clearance ranges, recommendations for adjusting dosing interval and/or dose are made to maintain similar area under the curve ranges over a 24-hour period.

Thus the recommended dosing frequency for $Cl_{cr} > 50$ ml/min/1.73 $m^2$, every eight hours; $Cl_{cr}$ range 25 to 50 ml/min/1.73 $m^2$, every 12 hours; $Cl_{cr}$ range 10 to 25 ml/min/1.73 $m^2$, every 24 hours; and $Cl_{cr} < 10$ ml/min/1.73 $m^2$, one-half the usual recommended dose every 24 hours. Because these dosage recommendations have been developed from the observed relationship between $Cl_{cr}$ and $Cl_{tot}$ they need to be further validated in future clinical trials.

In conclusion, the extensive pharmacokinetic evaluation of acyclovir during Phase I and II safety and efficacy trials has provided disposition parameters that have been useful in the design of other clinical protocols and has provided useful information to aid in developing dosage guidelines for the anticipated general use of the drug. Although these early studies have been informative, certain important information such as the relationship between plasma levels and efficacy remains to be defined in future studies.

## REFERENCES


1. Quinn RP, de Miranda P, Gerald L, Good SS. A sensitive radioimmunoassay for the antiviral agent BW 248U (9-(2-hydroxyethoxymethyl) guanine). Anal Biochem 1979; 98: 319–328.
2. Metzler CM, Elfring GK, McEwen AJ: A package of computer programs for pharmacokinetic modeling. Biometrics 1974; 30: 562–563.
3. de Miranda P, Whitley RJ, Blum MR, et al.: Acyclovir kinetics after intravenous infusion. Clin Pharmacol Ther 1979; 26: 718–728.
4. Mawer GE, Lucas SB, Knowles BR, Stirland RM, Tooth JA: Computer-assisted prescribing of kanamycin for patients with renal insufficiency. Lancet 1972; 1: 12.
5. de Miranda P, Good SS, Laskin OL, Krasny HC, Connor JD, Lietman PS: Disposition of intravenous radioactive acyclovir. Clin Pharmacol Ther 1981; 30: 662–672.
6. Whitley RJ, Blum MR, Barton N, de Miranda P: Pharmacokinetics of acyclovir in humans following intravenous administration. A model for the development of parenteral antivirals. In this supplement; 165–171.
7. Brigden D, Bye A, Fowle ASE: Human pharmacokinetics of acyclovir (an antiviral agent) following rapid intravenous injection. J Antimicrob Chemother 1981; 7: 399–404.
8. Hallynck TH, Soep HH, Thomas JA, Boelaert J, Daneels R, Dettli L: Should clearance be normalized to body surface or to lean body mass? Br J Clin Pharmacol 1981; 11: 523–526.
9. Procedure "Fit Multiple." Prophet Statistics, Belt Beranek and Newman, Inc., Cambridge, Mass., April 1981.
10. Bennett WM, Singer I, Coggins CH: Guide to drug usage in adult patients with impaired renal function, a supplement. JAMA 1973; 223: 991–997.
11. Diem K, Lentner C, eds.: Scientific tables. Ardsley, New York: Ciba-Geigy, 1970.
12. Laskin OL, Longstreth JA, Whelton A, et al.: Effect of renal failure on the pharmacokinetics of acyclovir. In this supplement; 197–201.